

BACKEND

WEEK 1

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SEC 7

1.Create Database

The screenshot shows the MySQL Workbench interface. The main editor window displays the following SQL code:

```
10 • create table members(  
11     member_id int auto_increment primary key,  
12     full_name varchar(100) not null,  
13     dept varchar(5) not null  
14 );  
15 • describe members;
```

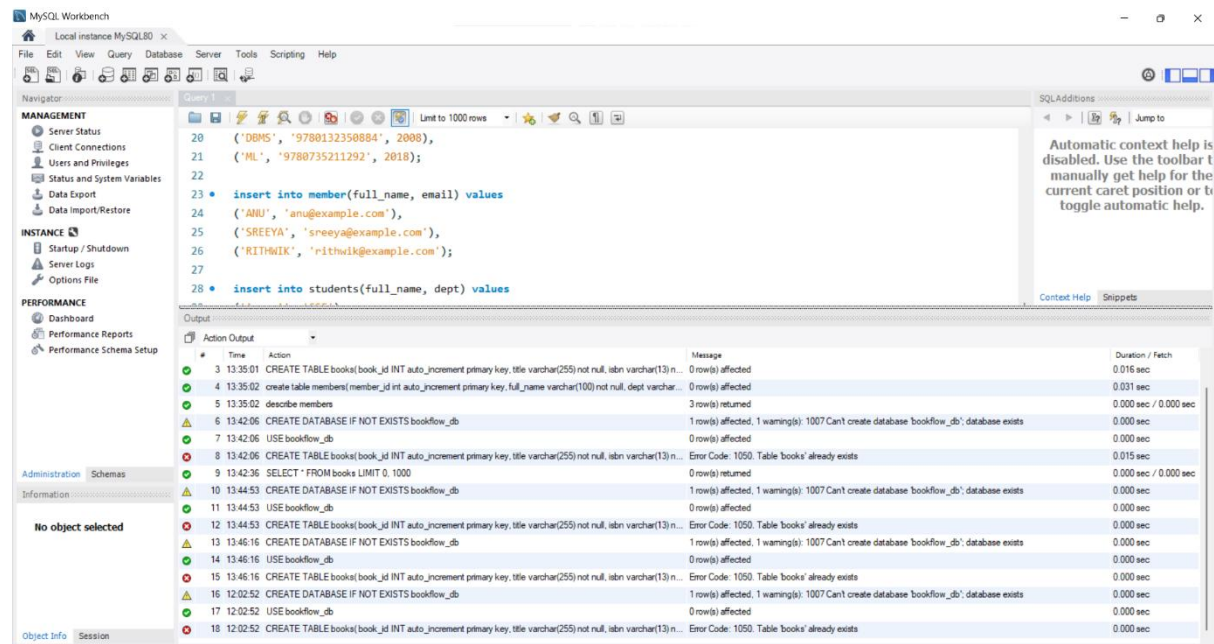
The Results window shows the output of the 'describe members' query:

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI		auto_increment
full_name	varchar(100)	NO			
dept	varchar(5)	NO			

The Action Output window shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	13:35:01	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected	0.015 sec
2	13:35:01	USE bookflow_db	0 row(s) affected	0.000 sec
3	13:35:01	CREATE TABLE books(book_id INT auto_increment primary key, title varchar(255) not null, isbn varchar(13) not null, year int(4) not null)	0 row(s) affected	0.016 sec
4	13:35:02	create table members(member_id int auto_increment primary key, full_name varchar(100) not null, dept varchar(5) not null)	0 row(s) affected	0.031 sec
5	13:35:02	describe members	3 row(s) returned	0.000 sec / 0.000 sec

2. Insert Records into Student_Marks Table



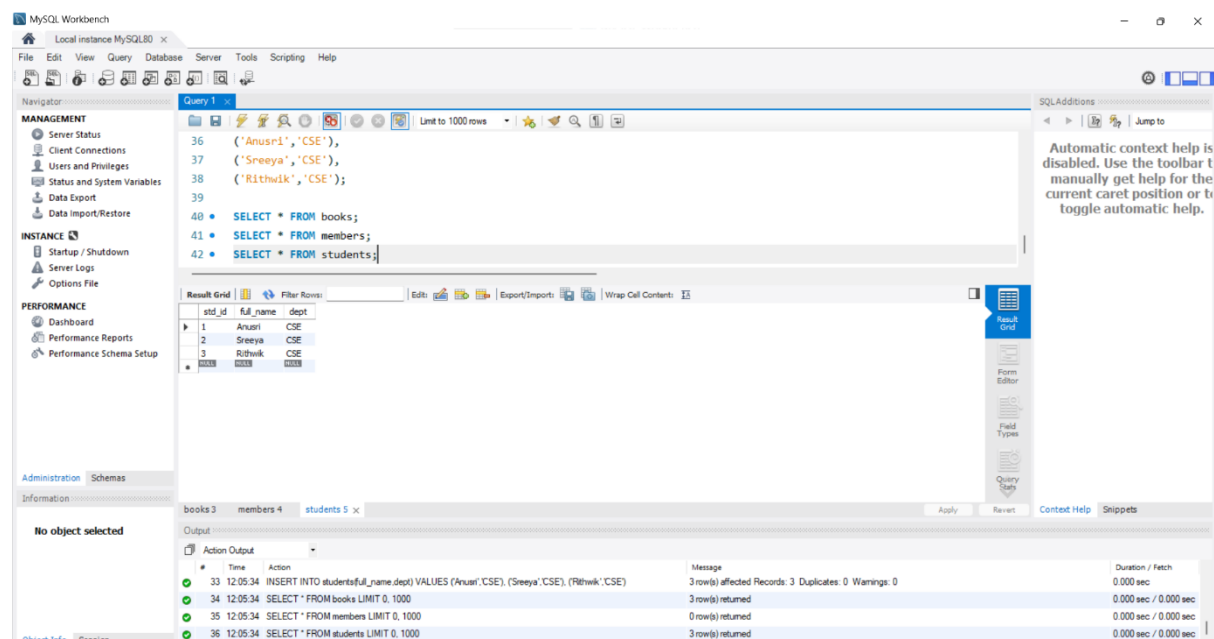
The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' tab with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. The 'INSTANCE' tab shows Startup/Shutdown, Server Logs, and Options File. The 'PERFORMANCE' tab shows Dashboard, Performance Reports, and Performance Schema Setup. The main editor area displays the following SQL queries:

```
20 ('DBMS', '9780132350884', 2008),
21 ('ML', '9780735211292', 2018);
22
23 • insert into member(full_name, email) values
24 ('ANU', 'anu@example.com'),
25 ('SREEYA', 'sreeya@example.com'),
26 ('RITHWIK', 'rithwik@example.com');
27
28 • insert into students(full_name, dept) values
```

The 'Output' tab shows the execution results of these queries. The 'Action Output' tab displays the following results:

#	Time	Action	Message	Duration / Fetch
3	13:35:01	CREATE TABLE books(book_id INT auto_increment primary key, title varchar(255) not null, isbn varchar(13) not null)	0 row(s) affected	0.016 sec
4	13:35:02	create table members(member_id int auto_increment primary key, full_name varchar(100) not null, dept varchar(10) not null)	0 row(s) affected	0.031 sec
5	13:35:02	describe members	3 row(s) returned	0.000 sec / 0.000 sec
6	13:42:06	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database exists	0.000 sec
7	13:42:06	USE bookflow_db	0 row(s) affected	0.000 sec
8	13:42:06	CREATE TABLE books(book_id INT auto_increment primary key, title varchar(255) not null, isbn varchar(13) not null)	Error Code: 1050. Table 'books' already exists	0.015 sec
9	13:42:36	SELECT * FROM books LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
10	13:44:53	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database exists	0.000 sec
11	13:44:53	USE bookflow_db	0 row(s) affected	0.000 sec
12	13:44:53	CREATE TABLE books(book_id INT auto_increment primary key, title varchar(255) not null, isbn varchar(13) not null)	Error Code: 1050. Table 'books' already exists	0.000 sec
13	13:46:16	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database exists	0.000 sec
14	13:46:16	USE bookflow_db	0 row(s) affected	0.000 sec
15	13:46:16	CREATE TABLE books(book_id INT auto_increment primary key, title varchar(255) not null, isbn varchar(13) not null)	Error Code: 1050. Table 'books' already exists	0.000 sec
16	12:02:52	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database exists	0.000 sec
17	12:02:52	USE bookflow_db	0 row(s) affected	0.000 sec
18	12:02:52	CREATE TABLE books(book_id INT auto_increment primary key, title varchar(255) not null, isbn varchar(13) not null)	Error Code: 1050. Table 'books' already exists	0.000 sec

```
30 • INSERT INTO books (title, isbn, published_year) VALUES
31 ('JAVA', '9780061122415', 1988),
32 ('DBMS', '9780132350884', 2008),
33 ('ML', '9780735211292', 2018);
34
35 • SELECT * FROM books;
```



The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' tab with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. The 'INSTANCE' tab shows Startup/Shutdown, Server Logs, and Options File. The 'PERFORMANCE' tab shows Dashboard, Performance Reports, and Performance Schema Setup. The main editor area displays the following SQL queries:

```
36 ('Anusi', 'CSE'),
37 ('Sreeya', 'CSE'),
38 ('Rithwik', 'CSE');
39
40 • SELECT * FROM books;
41 • SELECT * FROM members;
42 • SELECT * FROM students;
```

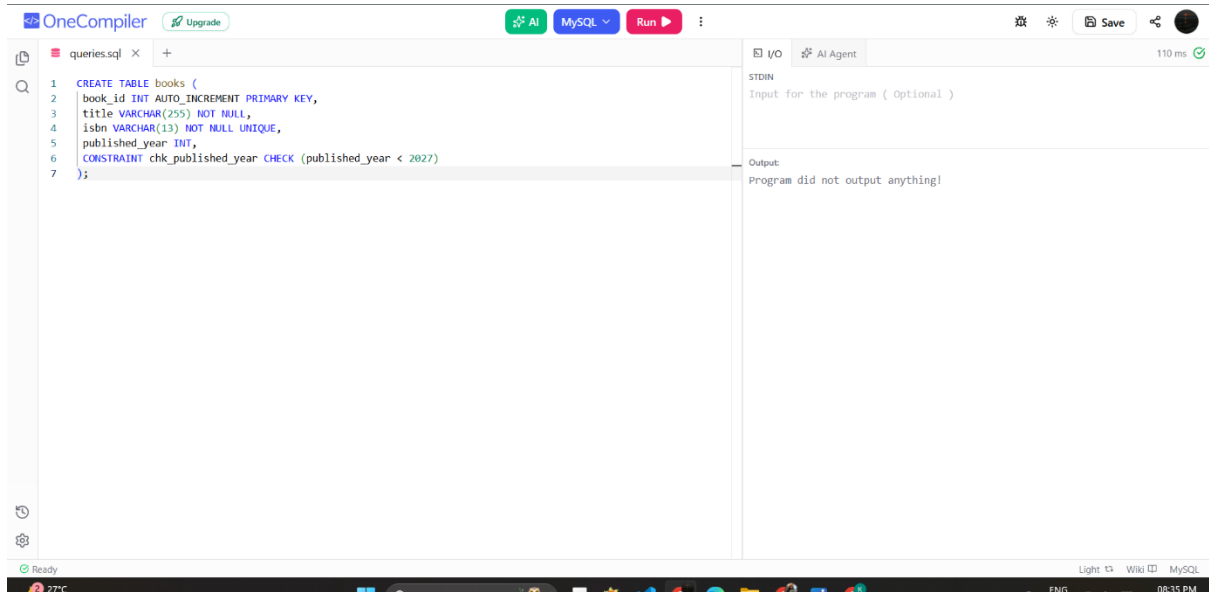
The 'Result Grid' tab shows the execution results of these queries. The 'books' table contains the following data:

std_id	full_name	dept
1	Anusi	CSE
2	Sreeya	CSE
3	Rithwik	CSE

The 'Output' tab shows the execution results of these queries. The 'Action Output' tab displays the following results:

#	Time	Action	Message	Duration / Fetch
33	12:05:34	INSERT INTO students(full_name,dept) VALUES ('Anusi','CSE'), ('Sreeya','CSE'), ('Rithwik','CSE')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
34	12:05:34	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
35	12:05:34	SELECT * FROM members LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
36	12:05:34	SELECT * FROM students LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

3. Display Student_Marks Table

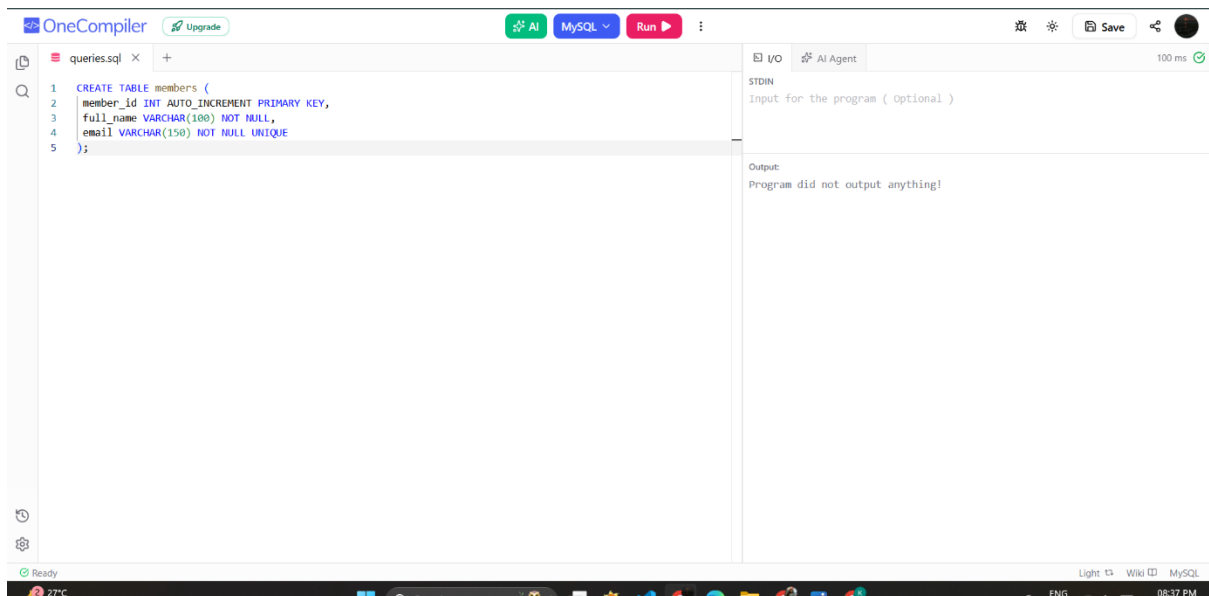


The screenshot shows the OneCompiler MySQL interface. The editor contains a SQL query to create a table named 'books'. The query is as follows:

```
1 CREATE TABLE books (  
2   book_id INT AUTO_INCREMENT PRIMARY KEY,  
3   title VARCHAR(255) NOT NULL,  
4   isbn VARCHAR(13) NOT NULL UNIQUE,  
5   published_year INT,  
6   CONSTRAINT chk_published_year CHECK (published_year < 2027)  
7 );
```

The right-hand pane shows the 'Output' section with the message: "Program did not output anything!". The top status bar indicates the execution time as 110 ms.

4. Count Total Students

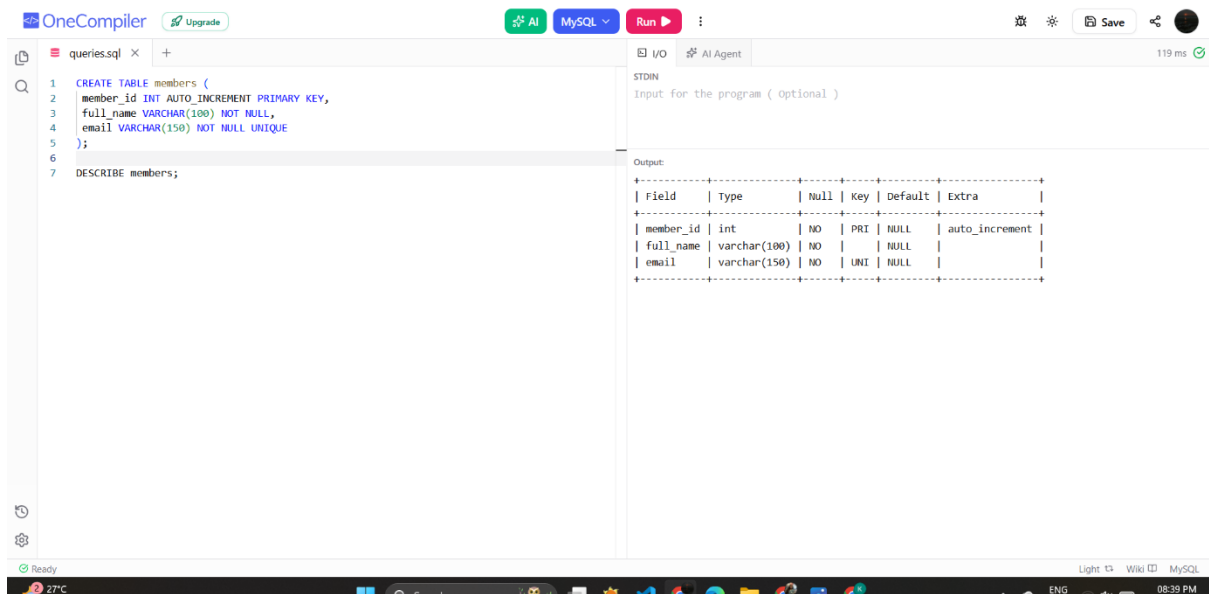


The screenshot shows the OneCompiler MySQL interface. The editor contains a SQL query to create a table named 'members'. The query is as follows:

```
1 CREATE TABLE members (  
2   member_id INT AUTO_INCREMENT PRIMARY KEY,  
3   full_name VARCHAR(100) NOT NULL,  
4   email VARCHAR(150) NOT NULL UNIQUE  
5 );
```

The right-hand pane shows the 'Output' section with the message: "Program did not output anything!". The top status bar indicates the execution time as 100 ms.

5. Calculate Total Marks



The screenshot shows the OneCompiler MySQL IDE interface. The editor on the left contains the following SQL code:

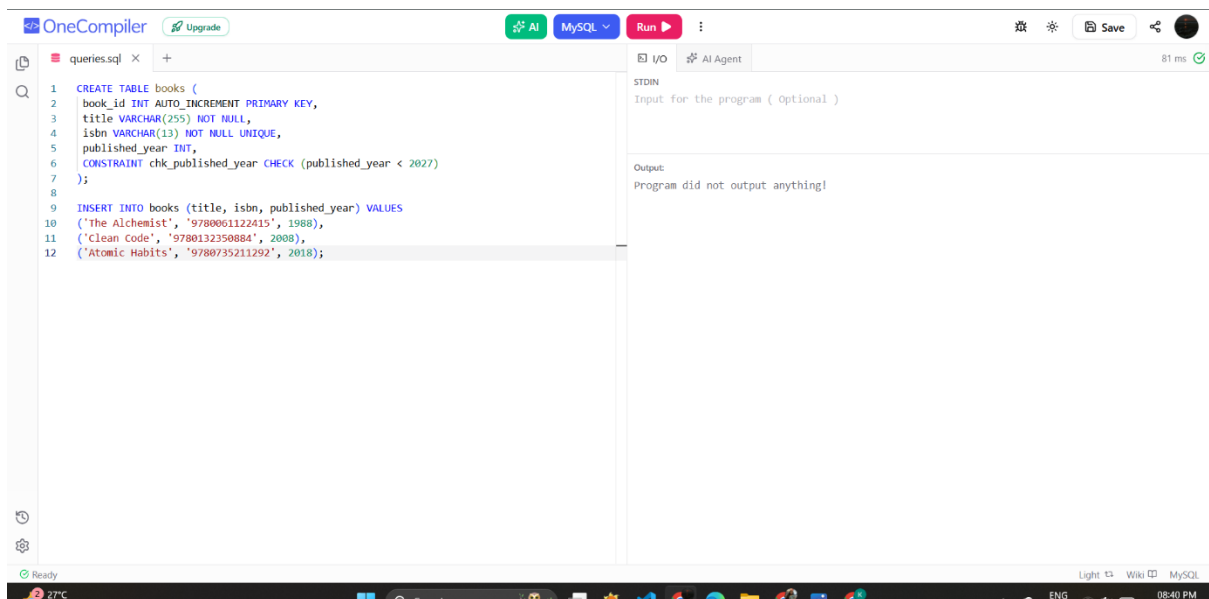
```
1 CREATE TABLE members (  
2   member_id INT AUTO_INCREMENT PRIMARY KEY,  
3   full_name VARCHAR(100) NOT NULL,  
4   email VARCHAR(150) NOT NULL UNIQUE  
5 );  
6  
7 DESCRIBE members;
```

The right-hand pane displays the output of the `DESCRIBE` command, showing the table's structure:

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

The status bar at the bottom indicates the system is ready, the temperature is 27°C, and the time is 08:39 PM.

6. Calculate Average Marks



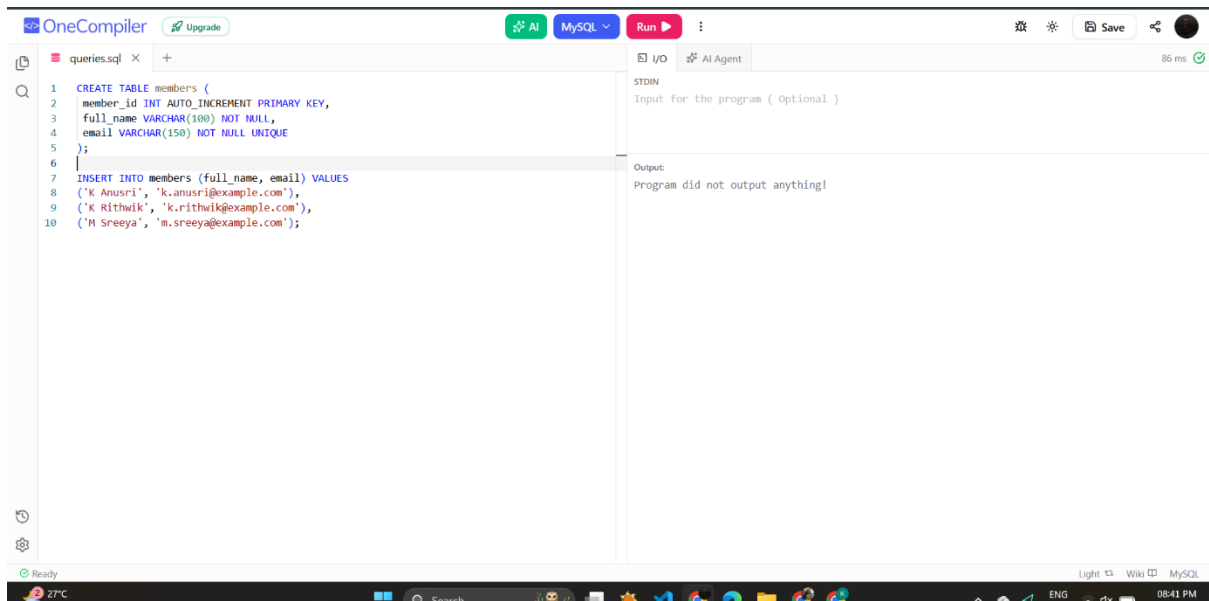
The screenshot shows the OneCompiler MySQL IDE interface. The editor on the left contains the following SQL code:

```
1 CREATE TABLE books (  
2   book_id INT AUTO_INCREMENT PRIMARY KEY,  
3   title VARCHAR(255) NOT NULL,  
4   isbn VARCHAR(13) NOT NULL UNIQUE,  
5   published_year INT,  
6   CONSTRAINT chk_published_year CHECK (published_year < 2027)  
7 );  
8  
9 INSERT INTO books (title, isbn, published_year) VALUES  
10 ('The Alchemist', '9780061122415', 1988),  
11 ('Clean Code', '9780132350884', 2008),  
12 ('Atomic Habits', '9780735211292', 2018);
```

The right-hand pane displays the output, which states: "Program did not output anything!".

The status bar at the bottom indicates the system is ready, the temperature is 27°C, and the time is 08:40 PM.

7. Find Maximum Marks

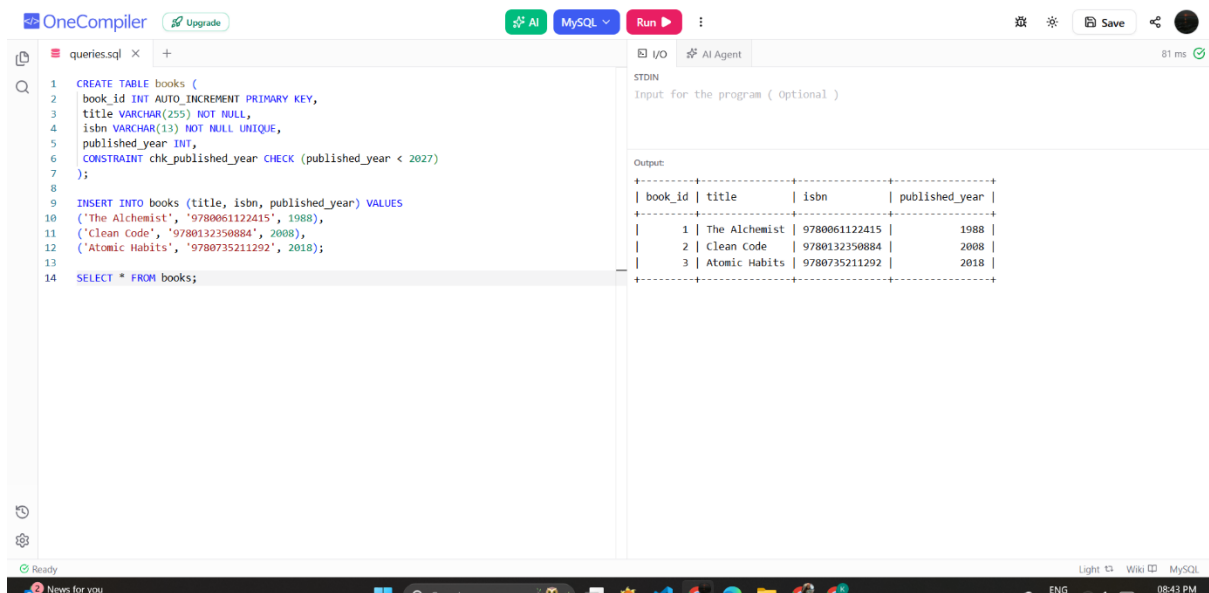


The screenshot shows the OneCompiler MySQL interface. The query editor contains the following SQL code:

```
1 CREATE TABLE members (  
2   member_id INT AUTO_INCREMENT PRIMARY KEY,  
3   full_name VARCHAR(100) NOT NULL,  
4   email VARCHAR(150) NOT NULL UNIQUE  
5 );  
6  
7 INSERT INTO members (full_name, email) VALUES  
8 ('K Anusri', 'k.anusri@example.com'),  
9 ('K Rithwik', 'k.rithwik@example.com'),  
10 ('M Sreeya', 'm.sreeya@example.com');
```

The output pane shows the message: "Program did not output anything!".

8. Find Minimum Marks



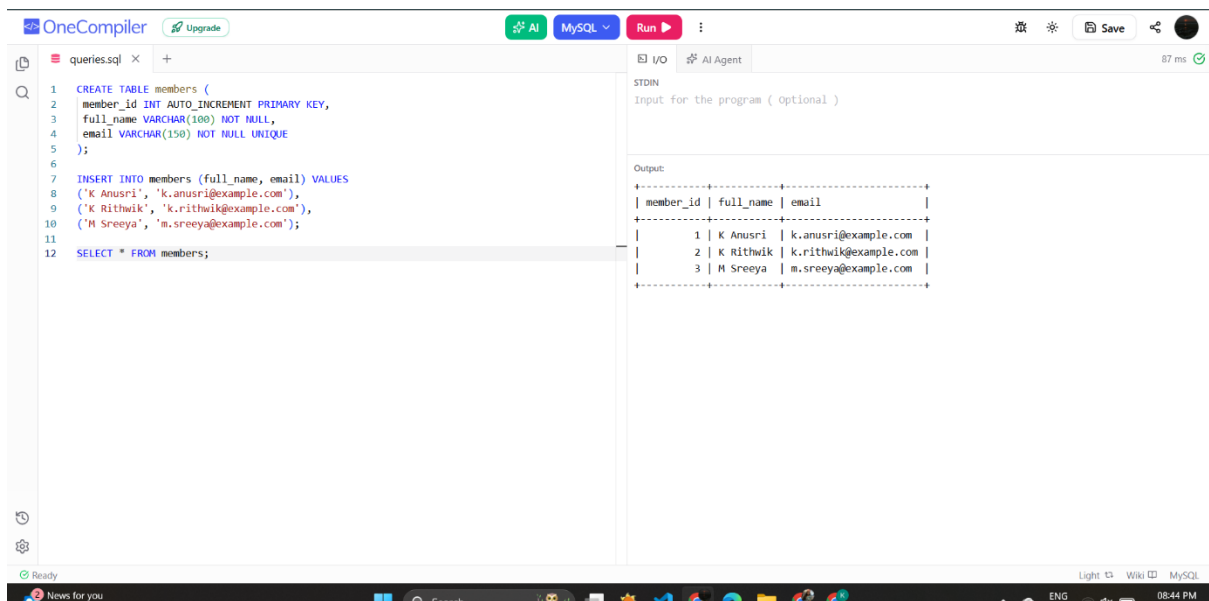
The screenshot shows the OneCompiler MySQL interface. The query editor contains the following SQL code:

```
1 CREATE TABLE books (  
2   book_id INT AUTO_INCREMENT PRIMARY KEY,  
3   title VARCHAR(255) NOT NULL,  
4   isbn VARCHAR(13) NOT NULL UNIQUE,  
5   published_year INT,  
6   CONSTRAINT chk_published_year CHECK (published_year < 2027)  
7 );  
8  
9 INSERT INTO books (title, isbn, published_year) VALUES  
10 ('The Alchemist', '9780061122415', 1988),  
11 ('Clean Code', '9780132350884', 2008),  
12 ('Atomic Habits', '9780735211292', 2018);  
13  
14 SELECT * FROM books;
```

The output pane shows the following table:

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

9. Display Students with Marks Greater than 85



The screenshot shows the OneCompiler MySQL interface. The left pane contains the following SQL code:

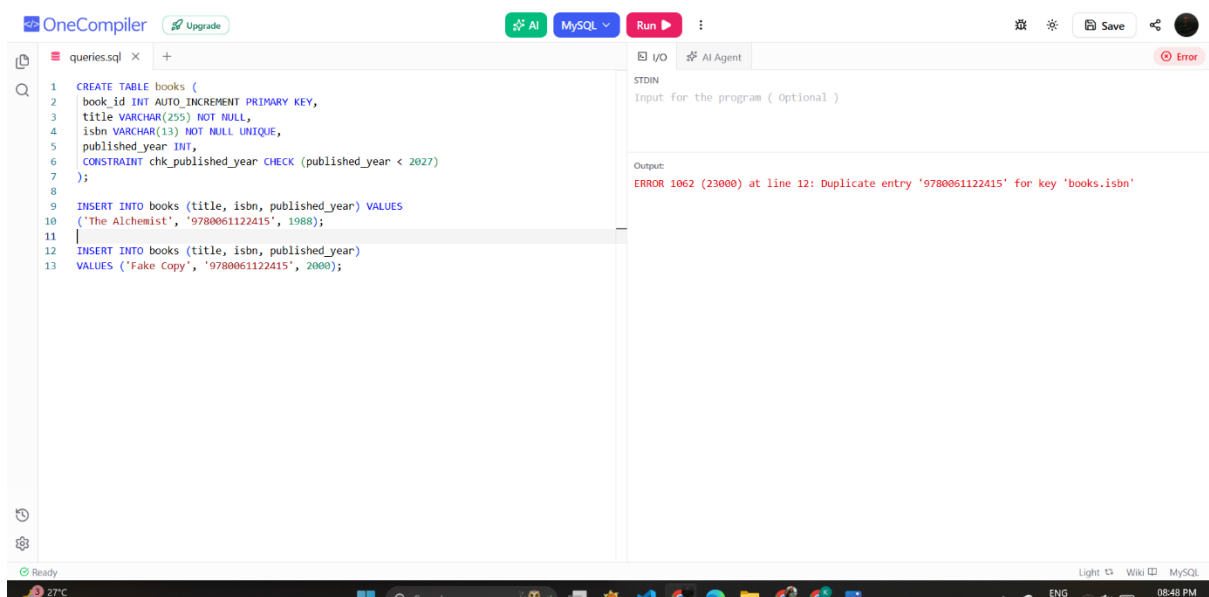
```
1 CREATE TABLE members (  
2   member_id INT AUTO_INCREMENT PRIMARY KEY,  
3   full_name VARCHAR(100) NOT NULL,  
4   email VARCHAR(150) NOT NULL UNIQUE  
5 );  
6  
7 INSERT INTO members (full_name, email) VALUES  
8 ('K Anusri', 'k.anusri@example.com'),  
9 ('K Rithwik', 'k.rithwik@example.com'),  
10 ('M Sreeya', 'm.sreeya@example.com');  
11  
12 SELECT * FROM members;
```

The right pane shows the output of the query, which is a table with 3 columns: member_id, full_name, and email. The output is as follows:

member_id	full_name	email
1	K Anusri	k.anusri@example.com
2	K Rithwik	k.rithwik@example.com
3	M Sreeya	m.sreeya@example.com

The status bar at the bottom indicates the query was executed successfully in 87 ms.

10. Display Students with Marks Greater than or Equal to 90

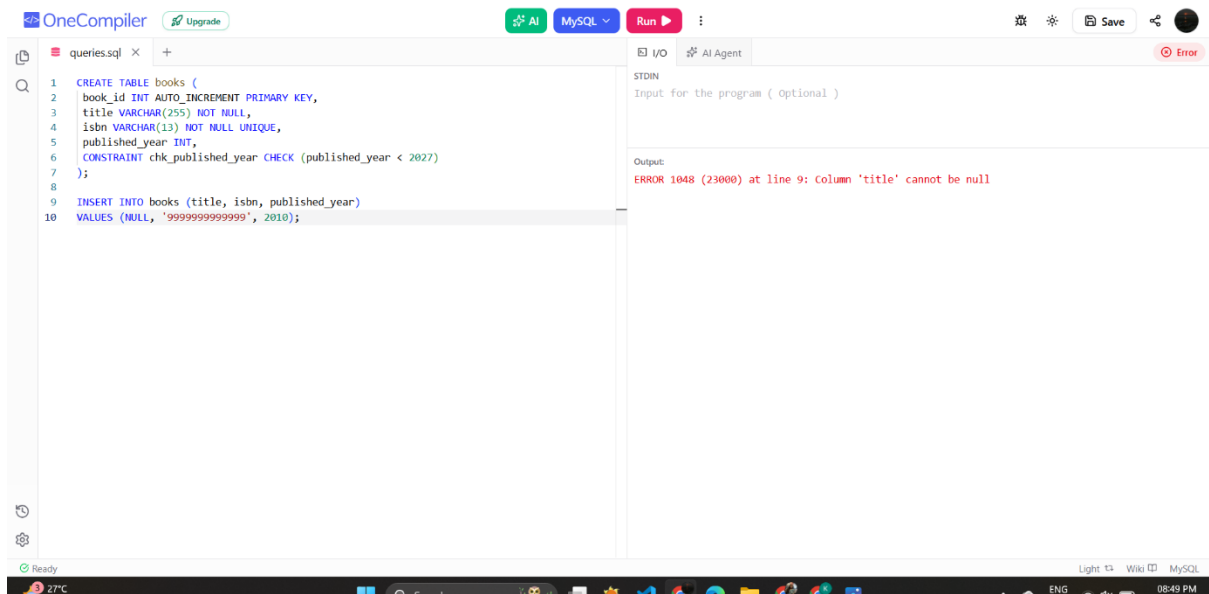


The screenshot shows the OneCompiler MySQL interface. The left pane contains the following SQL code:

```
1 CREATE TABLE books (  
2   book_id INT AUTO_INCREMENT PRIMARY KEY,  
3   title VARCHAR(255) NOT NULL,  
4   isbn VARCHAR(13) NOT NULL UNIQUE,  
5   published_year INT,  
6   CONSTRAINT chk_published_year CHECK (published_year < 2027)  
7 );  
8  
9 INSERT INTO books (title, isbn, published_year) VALUES  
10 ('The Alchemist', '9780061122415', 1988);  
11  
12 INSERT INTO books (title, isbn, published_year)  
13 VALUES ('Fake Copy', '9780061122415', 2000);
```

The right pane shows an error message: "ERROR 1062 (23000) at line 12: Duplicate entry '9780061122415' for key 'books.isbn'". The status bar at the bottom indicates the query failed.

11. Display Students with Marks Less than 80



The screenshot shows the OneCompiler MySQL IDE. The left pane contains a SQL script with the following code:

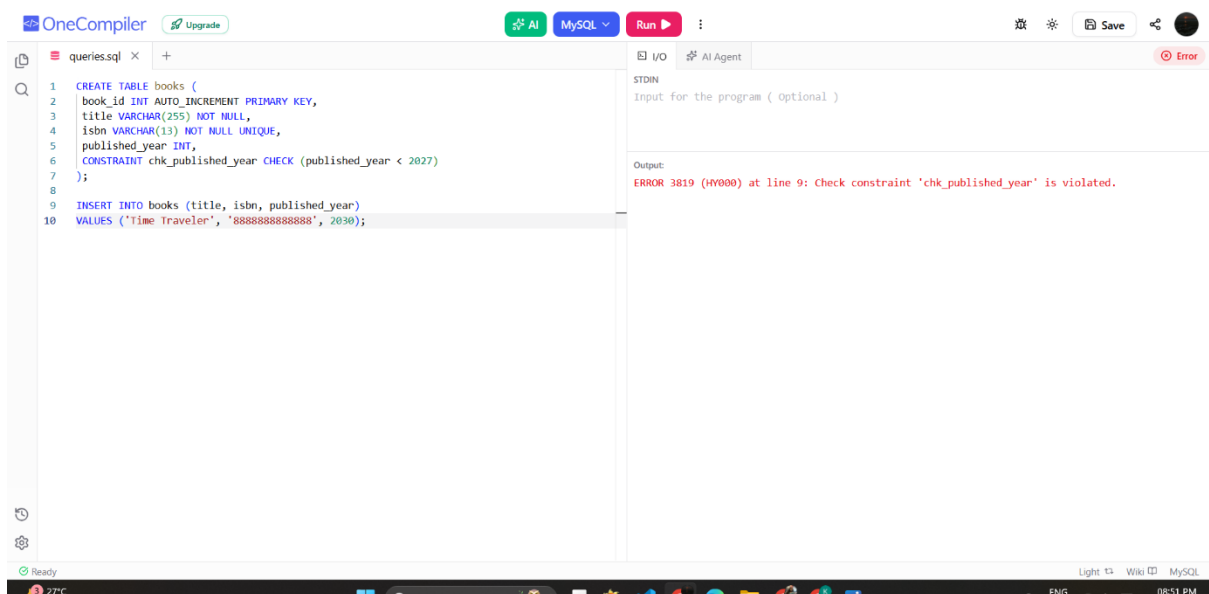
```
1 CREATE TABLE books (  
2   book_id INT AUTO_INCREMENT PRIMARY KEY,  
3   title VARCHAR(255) NOT NULL,  
4   isbn VARCHAR(13) NOT NULL UNIQUE,  
5   published_year INT,  
6   CONSTRAINT chk_published_year CHECK (published_year < 2027)  
7 );  
8  
9 INSERT INTO books (title, isbn, published_year)  
10 VALUES (NULL, '9999999999999', 2010);
```

The right pane shows the output area with an error message:

```
Output:  
ERROR 1048 (23000) at line 9: Column 'title' cannot be null
```

The status bar at the bottom indicates the system is ready, the temperature is 27°C, and the time is 08:49 PM.

12. Display Students with Marks Between 80 and 90



The screenshot shows the OneCompiler MySQL IDE. The left pane contains a SQL script with the following code:

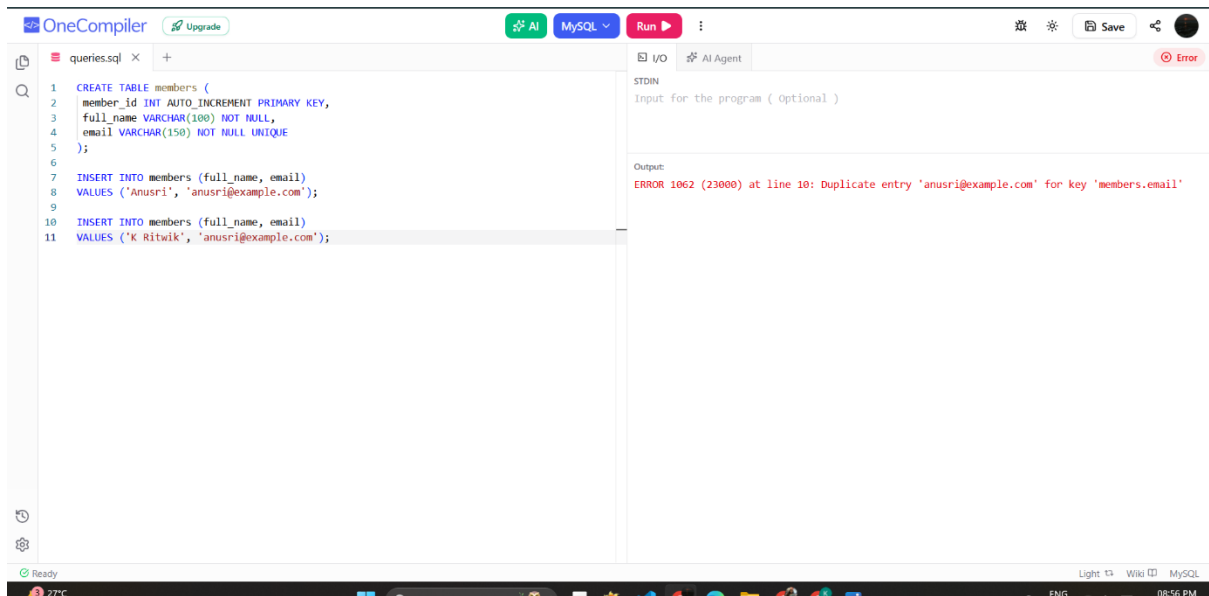
```
1 CREATE TABLE books (  
2   book_id INT AUTO_INCREMENT PRIMARY KEY,  
3   title VARCHAR(255) NOT NULL,  
4   isbn VARCHAR(13) NOT NULL UNIQUE,  
5   published_year INT,  
6   CONSTRAINT chk_published_year CHECK (published_year < 2027)  
7 );  
8  
9 INSERT INTO books (title, isbn, published_year)  
10 VALUES ('Time Traveler', '8888888888888', 2030);
```

The right pane shows the output area with an error message:

```
Output:  
ERROR 3819 (HY000) at line 9: Check constraint 'chk_published_year' is violated.
```

The status bar at the bottom indicates the system is ready, the temperature is 27°C, and the time is 08:51 PM.

13. Display Students Whose Name Starts with 'P'



The screenshot shows the OneCompiler MySQL interface. The SQL editor contains the following code:

```
1 CREATE TABLE members (  
2   member_id INT AUTO_INCREMENT PRIMARY KEY,  
3   full_name VARCHAR(100) NOT NULL,  
4   email VARCHAR(150) NOT NULL UNIQUE  
5 );  
6  
7 INSERT INTO members (full_name, email)  
8 VALUES ('Anusri', 'anusri@example.com');  
9  
10 INSERT INTO members (full_name, email)  
11 VALUES ('K Ritwik', 'anusri@example.com');
```

The output pane shows an error message: "ERROR 1062 (23000) at line 10: Duplicate entry 'anusri@example.com' for key 'members.email'".

14. Student Marks Table – SQL Queries and Output



The screenshot shows the OneCompiler MySQL interface. The SQL editor contains the following code:

```
1 CREATE TABLE student_marks (  
2   roll_no INT PRIMARY KEY,  
3   name VARCHAR(50),  
4   subject VARCHAR(50),  
5   marks DECIMAL(5,2)  
6 );  
7  
8 INSERT INTO student_marks (roll_no, name, subject, marks) VALUES  
9 (1, 'Ravi', 'Math', 85.50),  
10 (2, 'Sita', 'Math', 92.75),  
11 (3, 'Anil', 'Math', 78.40),  
12 (4, 'Priya', 'Math', 88.90),  
13 (5, 'Vijay', 'Math', 80.25),  
14 (6, 'subbusir', 'aws', 98.50),  
15 (7, 'dwaraka', 'dbms', 95.75),  
16 (8, 'ranjani', 'english', 97.40),  
17 (9, 'kavithamam', 'aws1', 99.90),  
18 (10, 'seetha', 'azure', 82.25);  
19  
20 SELECT * FROM student_marks;  
21  
22 SELECT COUNT(*) AS total_students FROM student_marks;  
23 SELECT SUM(marks) FROM student_marks;  
24 SELECT AVG(marks) FROM student_marks;  
25 SELECT MAX(marks) FROM student_marks;  
26 SELECT MIN(marks) FROM student_marks;  
27  
28 SELECT * FROM student_marks WHERE marks > 85;  
29 SELECT * FROM student_marks WHERE marks >= 90;  
30 SELECT * FROM student_marks WHERE marks < 80;  
31 SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;  
32 SELECT * FROM student_marks WHERE name LIKE 'P%';
```

The output pane shows the results of the queries. The first query returns a table with 10 rows of student data. The second query returns the total number of students (10). The third query returns the sum of marks (899.60).

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

total_students
10

SUM(marks)
899.60


```

53 SET SQL_SAFE_UPDATES = 0;
54 SELECT * FROM student_marks ORDER BY marks ASC;
55 SELECT * FROM student_marks ORDER BY marks DESC;
56 SELECT * FROM student_marks ORDER BY name ASC;
57 SELECT * FROM student_marks ORDER BY name DESC;
58
59 SELECT subject, SUM(marks) AS total_marks
60 FROM student_marks
61 GROUP BY subject;
62
63 SELECT subject, AVG(marks) AS avg_marks
64 FROM student_marks
65 GROUP BY subject;
66
67 SELECT subject, COUNT(*) AS num_students
68 FROM student_marks
69 GROUP BY subject;
70
71 SELECT subject, AVG(marks) AS avg_marks
72 FROM student_marks
73 GROUP BY subject
74 HAVING AVG(marks) > 90;
75
76 SELECT subject, COUNT(*) AS num_students
77 FROM student_marks
78 GROUP BY subject
79 HAVING COUNT(*) > 1;

```

Output:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

roll_no	name	subject	marks
2	Sita	Math	92.75
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90

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```

32 SELECT * FROM student_marks WHERE name LIKE 'PX';
33 SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
34 SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'PX');
35
36 SET SQL_SAFE_UPDATES = 0;
37
38 UPDATE student_marks
39 SET marks = 90.00
40 WHERE roll_no = 3;
41
42 UPDATE student_marks
43 SET subject = 'Remedial Math'
44 WHERE marks < 80;
45
46 DELETE FROM student_marks
47 WHERE roll_no = 5;
48
49 DELETE FROM student_marks
50 WHERE marks < 75;
51
52 SET SQL_SAFE_UPDATES = 1;
53
54 SELECT * FROM student_marks ORDER BY marks ASC;
55 SELECT * FROM student_marks ORDER BY marks DESC;
56 SELECT * FROM student_marks ORDER BY name ASC;
57 SELECT * FROM student_marks ORDER BY name DESC;
58
59 SELECT subject, SUM(marks) AS total_marks
60 FROM student_marks
61 GROUP BY subject;
62
63 SELECT subject, AVG(marks) AS avg_marks
64 FROM student_marks

```

Output:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90

Output:

```
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
|      3 | Anil | Math    | 78.40 |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
|      1 | Ravi  | Math    | 85.50 |
|      4 | Priya | Math    | 88.90 |
|      5 | Vijay | Math    | 80.25 |
|     10 | seetha | azure   | 82.25 |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
|      4 | Priya | Math    | 88.90 |
+-----+-----+-----+-----+
```

Output:

```
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
|      1 | Ravi  | Math    | 85.50 |
|      2 | Sita  | Math    | 92.75 |
|      5 | Vijay | Math    | 80.25 |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
|      1 | Ravi  | Math    | 85.50 |
|      2 | Sita  | Math    | 92.75 |
|      4 | Priya | Math    | 88.90 |
+-----+-----+-----+-----+
```

Output:

roll_no	name	subject	marks
10	seetha	azure	82.25
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
2	Sita	Math	92.75
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
6	subbusir	aws	98.50
9	kavithamam	aws1	99.90
roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
10	seetha	azure	82.25

Output:

roll_no	name	subject	marks
3	Anil	Math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	Priya	Math	88.90
8	ranjani	english	97.40
1	Ravi	Math	85.50
10	seetha	azure	82.25
2	Sita	Math	92.75
6	subbusir	aws	98.50
roll_no	name	subject	marks
6	subbusir	aws	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40
4	Priya	Math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	Anil	Math	90.00

Output:

+-----+		
subject	total_marks	
+-----+		
Math	357.15	
aws	98.50	
dbms	95.75	
english	97.40	
aws1	99.90	
azure	82.25	
+-----+		

+-----+		
subject	avg_marks	
+-----+		
Math	89.287500	
aws	98.500000	
dbms	95.750000	
english	97.400000	
aws1	99.900000	
azure	82.250000	
+-----+		

+-----+		
subject	num_students	
+-----+		
Math	4	
aws	1	
dbms	1	
english	1	
aws1	1	
azure	1	
+-----+		

+-----+-----+	
subject avg_marks	
+-----+-----+	
aws 98.500000	
dbms 95.750000	
english 97.400000	
aws1 99.900000	
+-----+-----+	
+-----+-----+	
subject num_students	
+-----+-----+	
Math 4	
+-----+-----+	